## 1. Demonstrate the use of Group by and Order by clause

- **Program :-**

```
CREATE TABLE EMP (
  EmpID NUMBER,
  Name VARCHAR2(20),
  DeptNo NUMBER,
  Salary NUMBER
);
INSERT INTO EMP VALUES (1, 'Ram', 10, 5000);
INSERT INTO EMP VALUES (2, 'Shyam', 20, 6000);
INSERT INTO EMP VALUES (3, 'Amit', 10, 7000);
INSERT INTO EMP VALUES (4, 'Neha', 20, 8000);
INSERT INTO EMP VALUES (5, 'Riya', 10, 6000);
select*from EMP;

SELECT DeptNo, SUM(Salary)
FROM EMP
GROUP BY DeptNo;

SELECT Name, Salary
FROM EMP
ORDER BY Salary DESC;
```

- **Output :-**

Table of EMP

| EMPID | NAME | DEPTNO | SALARY |
|---|---|---|---|
| 1 | Ram | 10 | 5000 |
| 2 | Shyam | 20 | 6000 |
| 3 | Amit | 10 | 7000 |
| 4 | Neha | 20 | 8000 |
| 5 | Riya | 10 | 6000 |

5 rows returned in 0.00 seconds CSV Export

**Group By**

| DEPTNO | SUM(SALARY) |
|---|---|
| 20 | 14000 |
| 10 | 18000 |

2 rows returned in 0.00 seconds

**Orde By**

| NAME | SALARY |
|---|---|
| Neha | 8000 |
| Amit | 7000 |
| Riya | 6000 |
| Shyam | 6000 |
| Ram | 5000 |

5 rows returned in 0.00 seconds

## 2. Create department table with the following structure

| Field Name | Data Type |
|---|---|
| Deptno | Int |
| DeptName | Varchar(30) |
| Location | Varchar(30) |

a) Add column designation to the department table.
b) Insert values into the table.
c) List the records of dept table grouped by deptno.
d) Update the record where deptno is 9.
e) Delete any column data from the table.

- **Program :-**

```
CREATE TABLE Department (
   Deptno INT PRIMARY KEY,


   DeptName VARCHAR(30),
   Location VARCHAR(30)
);
ALTER TABLE Department
ADD Designation VARCHAR(30);
INSERT INTO Department VALUES (1, 'HR', 'New York',
'Manager');
INSERT INTO Department VALUES (2, 'IT', 'London',
'Developer');
INSERT INTO Department VALUES (3, 'Finance', 'Delhi',
'Analyst');
SELECT Deptno, COUNT(*)
FROM Department
GROUP BY Deptno;
UPDATE Department
SET DeptName = 'Admin', Location = 'Mumbai'
WHERE Deptno = 9;
```

UPDATE Department

SET Designation = NULL;

1. Add column designation
2. Insert values
3. List records grouped by deptno
4. Update record where deptno is 9
5. Delete any column data (example: Designation data)

## ➢ Output :-

Results Explain Describe Saved SQL History

| DEPTNO | DEPTNAME | LOCATION | DESIGNATION |
|---|---|---|---|
| 1 | HR | New York | - |
| 2 | IT | London | - |
| 3 | Finance | Delhi | - |

3 rows returned in 0.00 seconds CSV Export

**3. Create department table with the following structure.**

| Field Name | Data Type |
|---|---|
| Deptno | Int |
| DeptName | Varchar(30) |
| Location | Varchar(30) |

**a) Add column designation to the department table.**

**b) Insert values into the table.**

**c) List the records of dept table grouped by deptno.**

**d) Update the record where deptno is 9.**

**e) Delete any column data from the table.**

- **Program :-**

```
CREATE TABLE Department2 (
   Deptno INT PRIMARY KEY,
   DeptName VARCHAR(30),
   Location VARCHAR(30)
);
ALTER TABLE Department2
ADD Designation VARCHAR(30);
INSERT INTO Department2 VALUES (10, 'Sales', 'Paris', 'Executive');
INSERT INTO Department2 VALUES (11, 'Support', 'Berlin', 'Assistant');
SELECT Deptno, COUNT(*)
FROM Department2
GROUP BY Deptno;
UPDATE Department2
SET DeptName = 'Operations'
WHERE Deptno = 9;
UPDATE Department2
SET Designation = NULL;
Select * from Department2;
```

- **Output :-**

Results Explain Describe Saved SQL History

| DEPTNO | DEPTNAME | LOCATION | DESIGNATION |
|---|---|---|---|
| 10 | Sales | Paris | - |
| 11 | Support | Berlin | - |

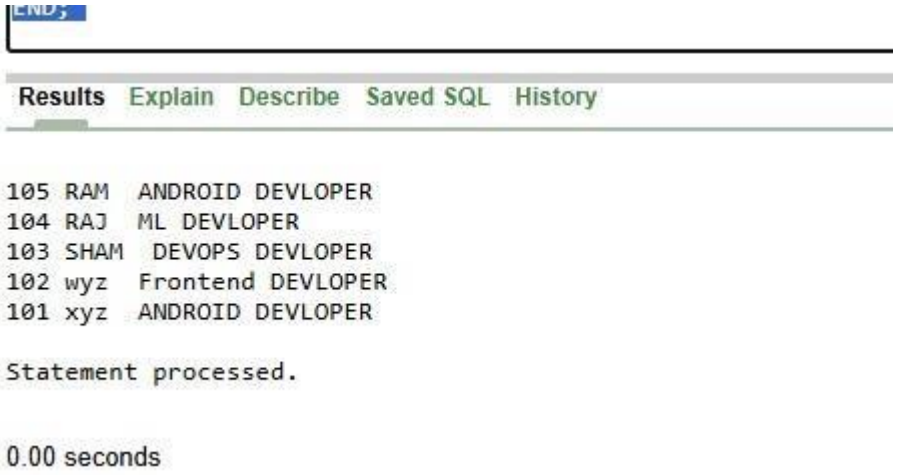
2 rows returned in 0.02 seconds CSV Export

## 4. Write PL/SQL code to display employee details using Explicit Cursors ?

- **Program :-**

```
CREATE TABLE EMPLOYEE2(empId int ,empName varchar(40),designation varchar(30));

INSERT INTO EMPLOYEE2 values (101,'xyz','ANDROID DEVLOPER');
INSERT INTO EMPLOYEE2 values (102,'wyz','Frontend DEVLOPER');
INSERT INTO EMPLOYEE2 values (103,'SHAM','DEVOPS DEVLOPER');
INSERT INTO EMPLOYEE2 values (104,'RAJ','ML DEVLOPER');
INSERT INTO EMPLOYEE2 values (105,'RAM','ANDROID DEVLOPER');

DECLARE
  CURSOR emp IS SELECT empId ,empName,designation  FROM
EMPLOYEE2;    v_empid EMPLOYEE.empId%TYPE;
v_empName EMPLOYEE.empName%TYPE;
  v_designation EMPLOYEE.designation%TYPE;


BEGIN
OPEN emp;
LOOP
  FETCH emp INTO v_empid,v_empname,v_designation;
     EXIT WHEN emp%NOTFOUND;
     DBMS_OUTPUT.PUT_LINE(v_empid || ' ' || v_empname || '  ' || v_designation);
 END LOOP;
CLOSE emp;

END;
```

- **Output :-**


END;
Results Explain Describe Saved SQL History
105 RAM ANDROID DEVLOPER
104 RAJ ML DEVLOPER
103 SHAM DEVOPS DEVLOPER
102 wyz Frontend DEVLOPER
101 xyz ANDROID DEVLOPER
Statement processed.
0.00 seconds

## 5. PL/SQL code to retrieve employee name, join date, and designation from employee database of an employee whose number is input by the user.

- **Program :-**

```
CREATE TABLE EMPLOYEE2(empId int ,empName varchar(40),designation varchar(30));

INSERT INTO EMPLOYEE2 values (101,'xyz','ANDROID DEVLOPER');
INSERT INTO EMPLOYEE2 values (102,'wyz','Frontend DEVLOPER');
INSERT INTO EMPLOYEE2 values (103,'SHAM','DEVOPS DEVLOPER');
INSERT INTO EMPLOYEE2 values (104,'RAJ','ML DEVLOPER');
INSERT INTO EMPLOYEE2 values (105,'RAM','ANDROID DEVLOPER');

DECLARE     v_empid
EMPLOYEE2.empId%TYPE := 101;
v_empname
EMPLOYEE2.empName%TYPE;
v_designation
EMPLOYEE2.designation%TYPE;

BEGIN
   SELECT empName, designation
   INTO v_empname, v_designation
   FROM EMPLOYEE2
   WHERE empId = v_empid;

   DBMS_OUTPUT.PUT_LINE('ID: ' || v_empid);
   DBMS_OUTPUT.PUT_LINE('Name: ' || v_empname);
   DBMS_OUTPUT.PUT_LINE('Designation: ' || v_designation);

EXCEPTION
```

```
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found');
    WHEN TOO_MANY_ROWS THEN
        DBMS_OUTPUT.PUT_LINE('More than one employee found');
END;
```

- **Output :-**

**6. Write a PL/SQL code to update the salary of employees who earn less than the average salary using cursor.**

- **Program :-**

```
BEGIN
   UPDATE EMPLOYEE2
   SET salary = salary + (salary * 0.10)
   WHERE salary < (SELECT AVG(salary) FROM EMPLOYEE2);

   COMMIT;
END;
/
SELECT * FROM EMPLOYEE2;
```

- **Output :-**

Results Explain Describe Saved SQL History

| EMPID | EMPNAME | DESIGNATION | SALARY |
|---|---|---|---|
| 105 | RAM | ANDROID DEVLOPER | 27500 |
| 104 | RAJ | ML DEVLOPER | 22000 |
| 103 | SHAM | DEVOPS DEVLOPER | 50000 |
| 102 | wyz | Frontend DEVLOPER | 40000 |
| 101 | xyz | ANDROID DEVLOPER | 33000 |

rows returned in 0.00 seconds CSV Export

**7. Write a row trigger to insert the existing values of the salary table into a new table when the salary table is updated.**

- **Program :-**

```
CREATE TABLE
SALARY_BACKUP (
empId NUMBER,
old_salary NUMBER,
updated_on DATE
);
CREATE OR REPLACE TRIGGER trg_salary_update
BEFORE UPDATE ON EMPLOYEE2
FOR EACH ROW
BEGIN
   INSERT INTO SALARY_BACKUP
   VALUES (:OLD.empId, :OLD.salary, SYSDATE);
END;
/
UPDATE EMPLOYEE2
SET salary = salary + 1000
WHERE empId = 101;

COMMIT;
SELECT * FROM SALARY_BACKUP;
```

- **Output :-**

| EMPID | OLD_SALARY | UPDATED_ON |
|---|---|---|
| 101 | 33000 | 31-MAR-26 |

1 rows returned in 0.01 seconds CSV Export

**8. Write a PL/SQL procedure to find the number of students ranging from 100–70%, 69–60%, 59– 50% & below 49% in each course from the STUDENT_COURSE table. The counts should be passed as parameters.**

- **Program :-**

```
CREATE TABLE
STUDENT_COURSE (     id
NUMBER,     percentage
NUMBER
);
INSERT INTO STUDENT_COURSE VALUES (1, 85);
INSERT INTO STUDENT_COURSE VALUES (2, 72);
INSERT INTO STUDENT_COURSE VALUES (3, 65);
INSERT INTO STUDENT_COURSE VALUES (4, 55);
INSERT INTO STUDENT_COURSE VALUES
(5, 45); INSERT INTO STUDENT_COURSE
VALUES (6, 30);

CREATE OR REPLACE PROCEDURE
student_count_proc(     a OUT NUMBER,     b
OUT NUMBER,     c OUT NUMBER,     d OUT
NUMBER
)
AS
BEGIN
   SELECT COUNT(*) INTO a FROM STUDENT_COURSE WHERE percentage >= 70;
   SELECT COUNT(*) INTO b FROM STUDENT_COURSE WHERE percentage >= 60
AND percentage < 70;
   SELECT COUNT(*) INTO c FROM STUDENT_COURSE WHERE percentage >= 50
AND percentage < 60;
   SELECT COUNT(*) INTO d FROM STUDENT_COURSE WHERE percentage < 50;
```

```
END;
/

DECLARE
  x
NUMBER;
y NUMBER;
z  NUMBER;
w NUMBER;
BEGIN
   student_count_proc(x, y, z, w);

   DBMS_OUTPUT.PUT_LINE('70-100: ' || x);
   DBMS_OUTPUT.PUT_LINE('60-69: ' || y);
   DBMS_OUTPUT.PUT_LINE('50-59: ' || z);
   DBMS_OUTPUT.PUT_LINE('Below 50: ' || w);
END;
/
```

- **Output :-**

**9. Create a store function that accepts 2 numbers and returns the addition of passed values. Also, write the code to call your function.**

- **Program :-**

```
CREATE OR REPLACE FUNCTION add_two_numbers(a NUMBER, b NUMBER)
RETURN NUMBER
IS
BEGIN
   RETURN a + b;
END;
/
DECLARE     result NUMBER;
BEGIN     result :=
add_two_numbers(10, 20);
   DBMS_OUTPUT.PUT_LINE('Sum = ' || result);
END;
/
```

- **Output :-**

```
Sum = 30

Statement processed.

0.00 seconds
```

**10. Write a PL/SQL function that accepts the department number and returns the total salary of the department. Also write a function call.**

- **Program :-**

```
CREATE TABLE
EMP_DEPT (     empId
NUMBER,     empName
VARCHAR2(40),     salary
NUMBER,
department_id NUMBER
);

INSERT INTO EMP_DEPT VALUES (101, 'A', 30000, 1);
INSERT INTO EMP_DEPT VALUES (102, 'B', 40000, 1);
INSERT INTO EMP_DEPT VALUES (103, 'C', 50000, 2);
INSERT INTO EMP_DEPT VALUES (104, 'D', 20000, 2);
INSERT INTO EMP_DEPT VALUES (105, 'E', 25000, 3);

CREATE OR REPLACE FUNCTION total_salary(dept_id NUMBER)
RETURN NUMBER
IS
   total NUMBER;
BEGIN
   SELECT SUM(salary) INTO total
   FROM EMP_DEPT
   WHERE department_id = dept_id;

   RETURN total;
END;
```

/

SELECT total_salary(1) FROM dual;

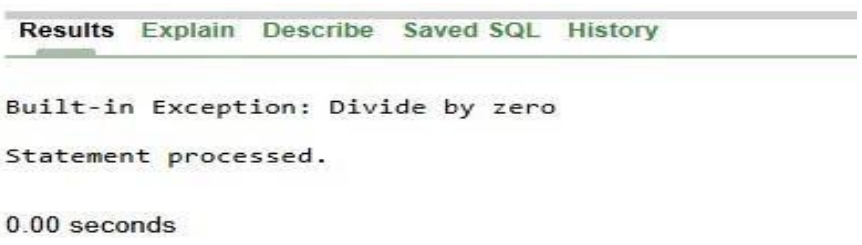
- **Output :-**

**11. Write a program to illustrate user-defined exceptions, built-in exceptions, and raise application error exceptions.**

- **Program :-**

```
DECLARE
   my_exception
EXCEPTION;     num
NUMBER := 0;     result
NUMBER; BEGIN
   result := 10 / num;
IF num = 0 THEN
      RAISE my_exception;
   END IF;
EXCEPTION
   WHEN ZERO_DIVIDE THEN
      DBMS_OUTPUT.PUT_LINE('Built-in Exception: Divide by
zero');     WHEN my_exception THEN
      DBMS_OUTPUT.PUT_LINE('User-defined Exception occurred');
   WHEN OTHERS THEN
      RAISE_APPLICATION_ERROR(-20001, 'Application Error occurred');
END;
/
```

- **Output :-**



## 12. Write a program to Reverse a String Using PL/SQL Block.

- **Program :-**

```
DECLARE
   str VARCHAR2(50) :=
'HELLO';     rev
VARCHAR2(50) := '';     i
NUMBER;
BEGIN
   FOR i IN REVERSE 1..LENGTH(str)
LOOP         rev := rev || SUBSTR(str, i,
1);
   END LOOP;

   DBMS_OUTPUT.PUT_LINE('Original: ' || str);
   DBMS_OUTPUT.PUT_LINE('Reversed: ' || rev);
END;
/
```

- **Output :-**

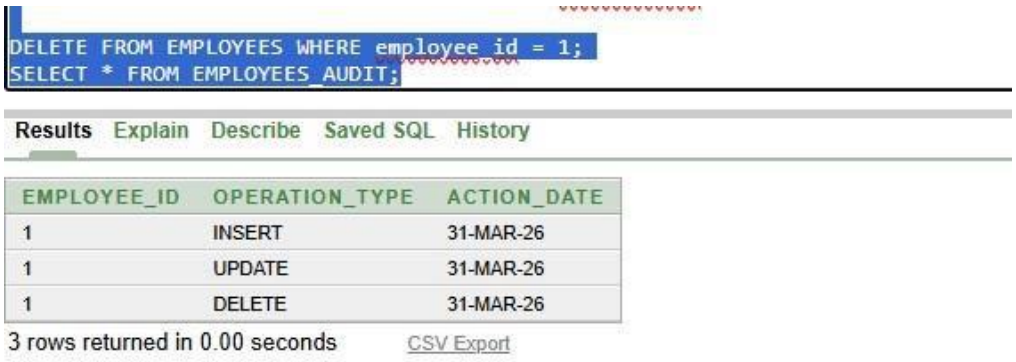
**13. Trigger for Auditing Table Changes Create a trigger that records changes to an EMPLOYEES table (INSERT, UPDATE, DELETE) into an EMPLOYEES_AUDIT table. Include details like EMPLOYEE_ID, OPERATION_TYPE, TIMESTAMP.**

- **Program :-**

```
CREATE TABLE
EMPLOYEES (
employee_id NUMBER,
name VARCHAR2(50),
salary NUMBER
);

CREATE TABLE
EMPLOYEES_AUDIT (
employee_id NUMBER,
operation_type VARCHAR2(10),
action_date DATE
);

CREATE OR REPLACE TRIGGER emp_audit_trigger
AFTER INSERT OR UPDATE OR DELETE ON EMPLOYEES
FOR EACH ROW
BEGIN
   IF INSERTING THEN
      INSERT INTO EMPLOYEES_AUDIT
      VALUES (:NEW.employee_id, 'INSERT', SYSDATE);

   ELSIF UPDATING THEN
      INSERT INTO EMPLOYEES_AUDIT
      VALUES (:NEW.employee_id, 'UPDATE', SYSDATE);
```

```
    ELSIF DELETING THEN
        INSERT INTO EMPLOYEES_AUDIT
        VALUES (:OLD.employee_id, 'DELETE', SYSDATE);
    END IF;
END;
/
INSERT INTO EMPLOYEES VALUES (1, 'A', 30000);

UPDATE EMPLOYEES SET salary = 35000 WHERE employee_id = 1;

DELETE FROM EMPLOYEES WHERE employee_id = 1;
SELECT * FROM EMPLOYEES_AUDIT;
```

- **Output :-**



## 14. Write a PL/SQL block to demonstrate the use of aggregate functions.

- **Program :-**

```
DECLARE
   total NUMBER;
avg_sal NUMBER;
max_sal NUMBER;
min_sal NUMBER;
count_emp NUMBER;
BEGIN
   SELECT
      COUNT(*),
      SUM(salary),
      AVG(salary),
      MAX(salary),
      MIN(salary)
INTO
count_emp,
      total,
avg_sal,
max_sal,
min_sal
   FROM EMP_DEPT;
   DBMS_OUTPUT.PUT_LINE('Count = ' || count_emp);
   DBMS_OUTPUT.PUT_LINE('Sum = ' || total);
   DBMS_OUTPUT.PUT_LINE('Average = ' || avg_sal);
   DBMS_OUTPUT.PUT_LINE('Max = ' || max_sal);
   DBMS_OUTPUT.PUT_LINE('Min = ' || min_sal);
END;
```

/

## i. Output :-


Results Explain Describe Saved SQL History
Count = 5
Sum = 165000
Average = 33000
Max = 50000
Min = 20000
Statement processed.
0.02 seconds

**15. Employee Bonus Calculation Using Cursor Write a PL/SQL program using an explicit cursor to calculate and display a 10% bonus for all employees whose salary is greater than 50,000. Assume a table EMPLOYEES with columns EMPLOYEE_ID, NAME, and SALARY.**

**a. Program :-**

```
DECLARE
   CURSOR emp_cursor IS
      SELECT empId, empName, salary FROM EMP_DEPT;

   v_id EMP_DEPT.empId%TYPE;
v_name
EMP_DEPT.empName%TYPE;
   v_salary EMP_DEPT.salary%TYPE;

BEGIN
   OPEN emp_cursor;

   LOOP
      FETCH emp_cursor INTO v_id, v_name, v_salary;
      EXIT WHEN emp_cursor%NOTFOUND;

      DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_name || ' ' || v_salary);
   END LOOP;

   CLOSE emp_cursor;
END;
/
```

## i. Output :-


Results Explain Describe Saved SQL History
101 A 30000
102 B 40000
103 C 50000
104 D 20000
105 E 25000
Statement processed.
0.00 seconds